

Exercise J01

Q1. The argument of the complex number $\sin \frac{6\pi}{5} + i \left(1 + \cos \frac{6\pi}{5}\right)$ is

- (A) $\frac{6\pi}{5}$ (B) $\frac{5\pi}{5}$ (C) $\frac{9\pi}{10}$ (D) $\frac{2\pi}{5}$

Ans. C

Sol. -

Q2. Number of complex numbers z satisfying $z^3 = \bar{z}$ is

- (A) 1 (B) 2 (C) 4 (D) 5

Ans. D

Sol. -

Q3. For $Z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}}$; $Z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}}$; $Z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}}$ which of the following holds good?

- (A) $\sum |Z_1|^2 = \frac{3}{2}$ (B) $\sum |Z_1|^4 + |Z_2|^4 = |Z_3|^{-8}$ (C) $\sum |Z_1|^3 + |Z_2|^3 = |Z_3|^{-6}$
(D) $\sum |Z_1|^4 + |Z_2|^4 = |Z_3|^8$

Ans. B

Sol. -

Q4. Let z and w are two non zero complex number such that $|z| = |w|$, and $\text{Arg}(z) + \text{Arg}(w) = \pi$ then -

- (A) $z = w$ (B) $z = \bar{w}$ (C) $\bar{z} = \bar{w}$ (D) $z = -\bar{w}$

Ans. D

Sol. -

Q5. If z and ω are two non- zero complex numbers such that $|z\omega| = 1$, and $\text{Arg}(z) - \text{Arg}(\omega) = \frac{\pi}{2}$, then $\bar{z}\omega$ is equal to-

- (A) $-i$ (B) 1 (C) -1 (D) i

Ans. A

Sol. -

Q6. If $\left(\frac{1+i}{1-i}\right)^x = 1$, then

- (A) $x = 2n + 1$, where n is any positive integer

(B) $x = 4n$, where n is any positive integer (C) $x = 2n$, where n is any positive integer

(D) $x = 4n + 1$, where n is any positive integer

Ans. B

Sol. -

Q7. If $z = x - iy$ and $z^{\frac{1}{3}} = p + iq$, then $\frac{\left(\frac{x}{p} + \frac{y}{q}\right)}{(p^2 + q^2)}$ is equal to-

(A) 1 (B) -1 (C) 2 (D) -2

Ans. D

Sol. $z = x - iy$

$$z^{\frac{1}{3}} = p + iq$$

$$z = p^3 - iq^3 + 3ip^2q - 3pq^2$$

By comparison

$$p^3 - 3pq^2 = x$$

$$\frac{x}{p} = p^2 - 3q^2$$

$$3p^2q - q^3 = -y$$

$$\frac{y}{q} = q^2 - 3p^2$$

$$\text{So that } \frac{\frac{x}{p} + \frac{y}{q}}{p^2 + q^2} = \frac{p^2 - 3q^2 + q^2 - 3p^2}{p^2 + q^2} = -2$$

Q8. If $|z + 4| \leq 3$, then the maximum and minimum value of $|z + 1|$ are -

(A) 4, 1 (B) 4, 0 (C) 6, 0 (D) 6, 1

Ans. C

Sol. -

Q9. Let α, β be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line

$\operatorname{Re}(z) = 1$, then it is necessary that :

(A) $|\beta| = 1$ (B) $\beta \in (1, \infty)$ (C) $\beta \in (0, 1)$ (D) $\beta \in (-1, 0)$

Ans. B

Sol. -

Q10. If z is a complex number of unit modulus and argument θ , then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals :

- (A) $-\theta$ (B) $\frac{\pi}{2} - \theta$ (C) θ (D) $\pi - \theta$

Ans. C

Sol. -

Q11. A complex number z is said to be unimodular if $|z| = 1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1 - 2z_2}{2 - \bar{z}_1 z_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies on a :

- (A) straight line parallel to x-axis (B) straight line parallel to y-axis
(C) circle of radius 2 (D) circle of radius $\sqrt{2}$

Ans. C

Sol. $(z_1 - 2z_2)(\bar{z}_1 - 2\bar{z}_2) = (2 - z_1\bar{z}_2)(2 - \bar{z}_1 z_2)$
 $z_1\bar{z}_1 - 2z_2\bar{z}_1 - 2z_1\bar{z}_2 + 4z_2\bar{z}_2 = 4 - 2z_1\bar{z}_2 + z_1\bar{z}_1 z_2\bar{z}_2$
 $|z_1|^2 |z_2|^2 - 4|z_2|^2 - |z_1|^2 + 4 = 0$
 $(|z_1|^2 - 4)(|z_2|^2 - 1) = 0$
 $|z_1|^2 - 4 = 0$
 $|z_1| = 2$
 Circle of radius 2

Q12. A value of θ for which $\frac{2 + 3i \sin \theta}{1 - 2i \sin \theta}$ is purely imaginary, is :

- (A) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. -

Q13. Let $z = \cos \theta + i \sin \theta$. Then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is

- (A) $\frac{2}{\sin 2^\circ}$ (B) $\frac{1}{3 \sin 2^\circ}$ (C) $\frac{1}{2 \sin 2^\circ}$ (D) $\frac{1}{4 \sin 2^\circ}$

Ans. D

Sol. -

Q14. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is

- (A) 48 (B) 32 (C) 40 (D) 80

Ans. A

Sol. -

Q15. The points z_1, z_2, z_3, z_4 in the complex plane are the vertices of a parallelogram taken in order if and only if :

- (A) $z_1 + z_4 = z_2 + z_3$ (B) $z_1 + z_3 = z_2 + z_4$ (C) $z_1 + z_2 = z_3 + z_4$ (D) None

Ans. B

Sol. -

Q16. The number of solutions of the system of equations $\operatorname{Re}(z^2) = 0, |z| = 2$ is

- (A) 4 (B) 3 (C) 2 (D) 1

Ans. A

Sol. -

Q17. If $z (\neq -1)$ is a complex number such that $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is equal to

- (A) 1 (B) 2 (C) 3 (D) 5

Ans. A

Sol. -

Q18. The curve represented by $|z| = \operatorname{Re}(z) + 2$ is

- (A) a straight line (B) a circle (C) an ellipse (D) a parabola

Ans. D

Sol. -

Q19. Let z_1 and z_2 be two nonreal complex cube roots of unity and $|z - z_1|^2 + |z - z_2|^2 = 1$ be the equation of a circle with z_1, z_2 as ends of a diameter then the value of λ is

- (A) 4 (B) 3 (C) 2 (D) $\sqrt{2}$

Ans. B

Sol. -

Q20. If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $\omega = \frac{2}{z}$, then the locus traced by ' ω ' in the complex plane is

- (A) $x - y - 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y + 1 = 0$ (D) $x + y + 1 = 0$

Ans. A

Sol. We have $|z - (1 + i)|^2 = 2$

$$\Rightarrow (x-1)^2 + (y-1)^2 = 2 \quad (\text{Put } z = x + iy)$$

$$\Rightarrow x^2 + y^2 = 2(x+y) \quad \dots\dots\dots(i)$$

$$\text{Let } \omega = h + ik = \frac{2}{z} = \frac{2}{x+iy} = \frac{2(x-iy)}{x^2+y^2}, \text{ so}$$

$$h = \frac{2x}{x^2+y^2}, k = \frac{-2y}{x^2+y^2}$$

$$\Rightarrow h - k = \frac{2(x+y)}{x^2+y^2} = 1 \quad (\text{from equation (1)})$$

$\therefore$ Locus of the point ω (h, k) will be $x - y = 1$

Q21. If $|z-2| \geq |z-4|$ then correct statement is-

- (A) $\text{Re}(z) \geq 3$ (B) $\text{Re}(z) \leq 3$ (C) $\text{Re}(z) \geq 2$ (D) $\text{Re}(z) \leq 2$

Ans. A

Sol. -

Q22. If $w = \frac{z}{z - \frac{1}{3}i}$ and $|w| = 1$, then z lies on -

- (A) an ellipse (B) a circle (C) a straight line (D) a parabola

Ans. C

Sol. -

Q23. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is

- (A) 0 (B) 2 (C) 7 (D) 17

Ans. B

Sol. -

Q24. If $Z \neq 1$ and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies :

- (A) On the imaginary axis
(B) Either on the real axis or on a circle passing through the origin
(C) On a circle with centre at the origin
(D) Either on the real axis or on a circle not passing through the origin

Ans. B

Sol. -

Q25. If z is a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$:

- (A) is strictly greater than $\frac{5}{2}$ (B) is strictly greater than $\frac{3}{2}$ but less than $\frac{5}{2}$
 (C) is equal to $\frac{5}{2}$ (D) lies in the interval $(1, 2)$

Ans. D

Sol. -

Q26. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on

- (A) a line not passing through the origin (B) $|z| = \sqrt{2}$ (C) the x-axis
 (D) the y-axis

Ans. D

Sol. -

Q27. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2 . The point z_2 is given by

- (A) $6 + 7i$ (B) $-7 + 6i$ (C) $7 + 6i$ (D) $-6 + 7i$

Ans. D

Sol. -

Q28. If z_1, z_2, z_3 are vertices of an equilateral triangle inscribed in the circle $|z| = 2$ and if $z_1 = 1 + i\sqrt{3}$, then

- (A) $z_2 = -2, z_3 = 1 + i\sqrt{3}$ (B) $z_2 = 2, z_3 = 1 - i\sqrt{3}$ (C) $z_2 = -2, z_3 = 1 + i\sqrt{3}$
 (D) $z_2 = 1 - i\sqrt{3}, z_3 = -1 - i\sqrt{3}$

Ans. C

Sol. -

Q29. If z_1 & $\bar{z}_1$ represent adjacent vertices of a regular polygon of n sides with centre at the origin and

if $\frac{\operatorname{Im} z_1}{\operatorname{Re} z_1} = \sqrt{2} - 1$ then the value of n is equal to

- (A) 8 (B) 12 (C) 16 (D) 24

Ans. A

Sol. -

- Q30. Let P denotes a complex number z on the Argand's plane, and Q denotes a complex number $\sqrt{2|z|^2} \text{cis} \left(\frac{\pi}{4} + \theta \right)$ where $\theta = \text{amp } z$. If 'O' is the origin, then the ΔOPQ is
- (A) isosceles but not right angled (B) right angled but not isosceles
(C) right isosceles (D) equilateral.

Ans. C

Sol. -

- Q31. z_1 & z_2 are two distinct points in an argand plane. If $a|z_1| = b|z_2|$, (where $a, b \in \mathbb{R}$) then the point $\frac{az_1}{bz_2} + \frac{bz_2}{bz_1}$ is a point on the :
- (A) line segment $[-2, 2]$ of the real axis
(B) line segment $[-2, 2]$ of the imaginary axis (C) unit circle $|z| = 1$
(D) the line with $\arg z = \tan^{-1} 2$.

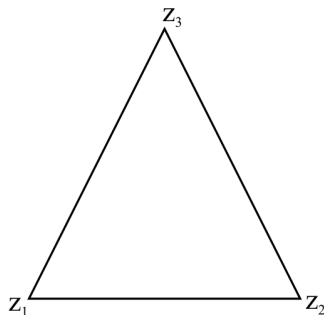
Ans. A

Sol. -

- Q32. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is
- (A) of area zero (B) right-angled isosceles (C) equilateral
(D) obtuse – angled isosceles

Ans. C

Sol.



$$\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$$

$$\arg \left(\frac{z_1 - z_3}{z_2 - z_3} \right) = \left(\frac{-\pi}{3} \right)$$

(-ve sign due to clockwise direction)

$$\left| \frac{z_1 - z_3}{z_2 - z_3} \right| = \left| \frac{1 - i\sqrt{3}}{2} \right|$$

$$\left| \frac{z_1 - z_3}{z_2 - z_3} \right| = 1 \Rightarrow |z_1 - z_3| = |z_2 - z_3|$$

Two sides are equal then corresponding angles are also equal.

Every angle of triangles is $\frac{\pi}{3}$, so that it's an equilateral triangle.

Q33. The vector $z = -4 + 5i$ is turned counter clockwise through an angle of 180° & stretched 1.5 times. The complex number corresponding to the newly obtained vector is :

- (A) $6 - \frac{15}{2}i$ (B) $-6 + \frac{15}{2}i$ (C) $6 + \frac{15}{2}i$ (D) None of these

Ans. A

Sol. -

Q34. Points z_1 & z_2 are adjacent vertices of a regular octagon. The vertex z_3 adjacent to z_2 ($z_3 \neq z_1$) is represented by :

- (A) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 + z_2)$ (B) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_1 - z_2)$ (C) $z_2 + \frac{1}{\sqrt{2}}(1 \pm i)(z_2 + z_1)$

(D) None of these

Ans. C

Sol. -

Q35. If $(\cos \theta + i \sin \theta)(\cos 2\theta + i \sin 2\theta) \dots (\cos n\theta + i \sin n\theta) = 1$, then the value of θ is

- (A) $4m\pi, m \in \mathbb{Z}$ (B) $\frac{2m\pi}{n(n+1)}, m \in \mathbb{Z}$ (C) $\frac{4m\pi}{n(n+1)}, m \in \mathbb{Z}$ (D) $\frac{m\pi}{n(n+1)}, m \in \mathbb{Z}$

Ans. C

Sol. -

Q36. The complex number ω satisfying the equation $\omega^3 = 8i$ and lying in the second quadrant on the complex plane is :

- (A) $-\sqrt{3} + i$ (B) $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$ (C) $-2\sqrt{3} + i$ (D) $-\sqrt{3} + 2i$

Ans. A

Sol. $\omega = 2 \cdot i^{1/3} = 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)^{1/3} = 2 \left[\cos \frac{2n\pi + \frac{\pi}{2}}{3} + i \sin \frac{2n\pi + \frac{\pi}{2}}{3} \right]$
 put $n = 1$ $= 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right] = -\sqrt{3} + i$ Ans.

Q37. If m and n are the smallest positive integers satisfying the relation

$$\left(2\cos \frac{\pi}{6} \right)^m = \left(4\cos \frac{\pi}{4} \right)^n, \text{ then } (m+n) \text{ has the value equal to}$$

- (A) 120 (B) 96 (C) 72 (D) 60

Ans. C

Sol. $2^{m-2n} \left[\cos \frac{m\pi}{6} + i \sin \frac{m\pi}{6} \right] = \left[\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right]$

for equality $m = 2n$

$$\frac{n\pi}{4} = \frac{m\pi}{6} + 2k\pi \quad k \in I$$

put $m = 2n$

$$\frac{n\pi}{4} = \frac{n\pi}{3} + 2k\pi \quad - \left(\frac{n\pi}{12} \right) = 2k\pi \quad (\text{ignore } (-)\text{ve sign})$$

$n = 24k$; $m = 48k$; for m, n to be smallest $m + n = 72$

Q38. Least positive argument of the 4th root of the complex number $2 - i\sqrt{12}$ is

- (A) $\pi/6$ (B) $5\pi/12$ (C) $7\pi/12$ (D) $11\pi/12$

Ans. B

Sol. -

Q39. If ω is an imaginary cube root of unity then $(1 + \omega - \omega^2)(1 + \omega^2 - \omega)$ equals-

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. -

Q40. If the cube roots of unity are $1, \omega, \omega^2$ then the roots of the equation $(x - 1)^3 + 8 = 0$, are -

- (A) $-1, -1 + 2\omega, -1 - 2\omega^2$ (B) $-1, -1, -1$ (C) $-1, 1 - 2\omega, 1 - 2\omega^2$
(D) $-1, 1 + 2\omega, 1 + 2\omega^2$

Ans. C

Sol. -

Q41. If $\omega (\neq 1)$ is a cube root of unity and $(1 + \omega)^7 = A + B\omega$. Then (A, B) equals :

- (A) $(1, 0)$ (B) $(-1, 1)$ (C) $(0, 1)$ (D) $(1, 1)$

Ans. D

Sol. -

Q42. Let ω be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$ then k is

equal to

- (A) -1 (B) 1 (C) -z (D) z

Ans. C

Sol. $\omega = \frac{-1}{2} + \frac{i\sqrt{3}}{2}, \quad 1 + \omega + \omega^2 = 0, \quad \omega^3 = 1$

$$3K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

then expand

$$3K = 3(\omega^2 - \omega)$$

$$= -3i\sqrt{3}$$

$$K = -i\sqrt{3} = -z$$

Q43. If $\alpha, \beta \in \mathbb{C}$ are the distinct roots, of the equation $x^2 - x + 1 = 0$, then $a^{101} + b^{107}$ is equal to

- (A) 1 (B) 2 (C) -1 (D) 0

Ans. A

Sol. -

Q44. The value of $\sum_{k=1}^{10} \left(\sin \frac{2k\pi}{11} + i \cos \frac{2k\pi}{11} \right)$ is -

- (A) 1 (B) -1 (C) -i (D) i

Ans. C

Sol. -

Exercise J02

Q1. Solve the following equations over $\mathbb{C}$ and express the result in the form $a + ib$, $a, b \in \mathbb{R}$.

(a) $ix^2 - 3x - 2i = 0$ (b) $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

Ans. (a) $-i, -2i$ (b) $\frac{3-5i}{5}$ or $-\frac{1+i}{2}$

Sol. -

Q2. Let z be a complex number such that $z \in \mathbb{C} \setminus \mathbb{R}$ and $\frac{1+z+z^2}{1-z+z^2} \in \mathbb{R}$, then prove that $|z| = 1$.

Ans. -

Sol. -

Q3. Prove the identity, $|1 - z_1 \bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

Ans. -

Sol. -

Q4. Prove the identity, $|1 + z_1 \bar{z}_2|^2 + |z_1 - z_2|^2 = (1 + |z_1|^2)(1 + |z_2|^2)$

Ans. -

Sol. -

Q5. (a) Find all non-zero complex numbers Z satisfying $\bar{z} = iz^2$.
(b) If the complex numbers $z_1, z_2, \dots, z_n$ lie on the unit circle $|z| = 1$ then show that

$$|z_1 + z_2 + \dots + z_n| = |z_1^{-1} + z_2^{-1} + \dots + z_n^{-1}|.$$

Ans. (a) $\frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}, i$

Sol. -

Q6. Let $z = (0, 1) \in \mathbb{C}$. Express $\sum_{k=0}^n z^k$ in terms of the positive integer n .

Ans.
$$\begin{cases} (1, 0) & \text{for } n = 4k \\ (1, 1) & \text{for } n = 4k + 1 \\ (0, 1) & \text{for } n = 4k + 2 \\ (0, 0) & \text{for } n = 4k + 3 \end{cases}$$

Sol. -

Q7. (a) Let Z is complex satisfying the equation, $z^2 - (3+i)z + m + 2i = 0$, where $m \in \mathbb{R}$. Suppose the equation has a real root, then find the value of m .

(b) a, b, c are real numbers in the polynomial, $P(Z) = 2Z^4 + aZ^3 + bZ^2 + cZ + 3$

If two roots of the equation $P(Z) = 0$ are 2 and i, then find the value of 'a'.

Ans. (b) $-11/2$

Sol. -

Q8. (a) Find the real values of x & y for which $z_1 = 9y^2 - 4 - 10ix$ and $z_2 = 8y^2 - 20i$ are conjugate complex of each other.

(b) Find the value of $x^4 - x^3 + x^2 + 3x - 5$ if $x = 2 + 3i$

Ans. (a) $[(-2, 2); (-2, -2)]$ (b) $-(77 + 108i)$

Sol. -

Q9. (a) Solve the following equation $z^2 - (3 - 2i)z = (5i - 5)$ expressing your answer in the form of $(a + ib)$.

(b) If $(1 - i)$ is a root of the equation, $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$, then find the other two roots.

Ans. (a) $z = (2 + i)$ or $(1 - 3i)$; (b) $z = 1$ or $2 - i$

Sol. -

Q10. (a) If $iZ^3 + Z^2 - Z + i = 0$, then find $|Z|$.

(b) Let z_1 and z_2 be two complex numbers such that $\left| \frac{z_1 - 2z_2}{2 - z_1\bar{z}_2} \right| = 1$ and $|z_2| \neq 1$, find $|z_1|$.

Ans. (a) 1, (b) 2

Sol. -

Q11. Show that the product,

$\left[1 + \left(\frac{1+i}{2} \right) \right] \left[1 + \left(\frac{1+i}{2} \right)^2 \right] \left[1 + \left(\frac{1+i}{2} \right)^{2^2} \right] \dots \left[1 + \left(\frac{1+i}{2} \right)^{2^n} \right]$ is equal to $\left(1 - \frac{1}{2^{2^n}} \right) (1+i)$ where $n \geq 2$.

Ans. -

Sol. -

Q12. Find the minimum value of the expression $E = |z|^2 + |z - 3|^2 + |z - 6i|^2$ (where $z = x + iy$, $x, y \in \mathbb{R}$).

Ans. 30

Sol. -

Q13. Let $A = \{a \in \mathbb{R} \mid \text{the equation } (1 + 2i)x^3 - 2(3 + i)x^2 + (5 - 4i)x + 2a^2 = 0 \text{ has at least one real root. Find the value of } \sum_{a \in A} a^2.$

Ans. 18

Sol. -

Q14. One root of the cubic $2z^3 - (5 + 6i)z^2 + 9iz + 1 - 3i = 0$ is real. If all the three roots of this cubic are plotted on the complex plane, then find the area of the triangle formed by them.

Ans. $\frac{1}{4}$

Sol. -

Q15. Prove that $\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right) = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$. Hence deduce that

$$\left(1 + \sin \frac{\pi}{5} + i \cos \frac{\pi}{5} \right)^5 + i \left(1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5} \right)^5 = 0$$

Ans. -

Sol. -

Q16. Dividing $f(z)$ by $z - i$, we get the remainder i and dividing it by $z + i$, we get the remainder $1 + i$. Find the remainder upon the division of $f(z)$ by $z^2 + 1$.

Ans. $\frac{iz}{2} + \frac{1}{2} + i$

Sol. -

Q17. If z_1 and z_2 are two complex numbers such that $|z_1| < 1 < |z_2|$ then prove that

$$\left| \frac{1 - z_1 \bar{z}_2}{z_1 - z_2} \right| < 1.$$

Ans. -

Sol. -

Q18. Among the complex numbers z satisfying the condition $|n + 3 - \sqrt{3}i| = \sqrt{3}$, find the number having the least positive argument.

Ans. $-\frac{3}{2} + \frac{3\sqrt{3}}{2}i$

Sol. -

Q19. Let $z_1 = 1 + i$ and $z_2 = -1 - i$. Find $z_3 \in \mathbb{C}$ such that triangle z_1, z_2, z_3 is equilateral.

Ans. $z_3 = \sqrt{3}(1 - i)$ and $z'_3 = \sqrt{3}(-1 + i)$

Sol. -

Q20. Find the Cartesian equation of the locus of ' z ' in the complex plane satisfying, $|z - 4| + |z + 4| = 16$.

Ans. $\frac{x^2}{64} + \frac{y^2}{48} = 1$

Sol. -

Q21. Interpret the following locii in $z \in \mathbb{C}$.

(a) $1 < |z - 2i| < 3$ (b) $\operatorname{Re} \left(\frac{z + 2i}{iz + 2} \right) \leq 4 (z \neq 2i)$

(c) $\operatorname{Arg}(z + i) - \operatorname{Arg}(z - i) = \pi/2$ (d) $\operatorname{Arg}(z - a) = \pi/3$ where $a = 3 + 4i$.

Ans. (a) The region between the co centric circles with centre at $(0, 2)$ & radii 1 & 3 units

(b) region outside or on the circle with centre $\frac{1}{2} + 2i$ and radius $\frac{1}{2}$.

(c) semi circle (in the 1st & 4th quadrant) $x^2 + y^2 = 1$

(d) a ray emanating from the point $(3 + 4i)$ directed away from the origin & having equation $\sqrt{3}x - y + 4 - 3\sqrt{3} = 0$

Sol. -

Q22. Let $A \equiv z_1$; $B \equiv z_2$; $C \equiv z_3$ are three complex numbers denoting the vertices of an acute angled triangle. If the origin 'O' is the orthocentre of the triangle, then prove that

$$z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_3 \bar{z}_1 + \bar{z}_3 z_1$$

hence show that the ΔABC is a right angled triangle $\Leftrightarrow z_1 \bar{z}_2 + \bar{z}_1 z_2 = z_2 \bar{z}_3 + \bar{z}_2 z_3 = z_3 \bar{z}_1 + \bar{z}_3 z_1 = 0$

Ans. -

Sol. -

Q23. (a) Let $z = x + iy$ be a complex number, where x and y are real numbers. Let A and B be the sets defined by $A = \{z \mid |z| \leq 2\}$ and $B = \{z \mid (1 - i)z + (1 + i)\bar{z} \geq 4\}$. Find the area of the region $A \cap B$.

(b) For all real numbers x , let the mapping $f(x) = \frac{1}{x - i}$, where $i = \sqrt{-1}$. If there exist real number a, b, c and d for which $f(a), f(b), f(c)$ and $f(d)$ form a square on the complex plane. Find the area of the square.

Ans. (a) $\pi - 2$; (b) $1/2$

Sol. -

Q24. Find centre and radius of the circle determined by all complex numbers $z = x + iy$ satisfying $\left| \frac{(z - \alpha)}{(z - \beta)} \right| = k$, where $\alpha = \alpha_1 + i\alpha_2, \beta = \beta_1 + i\beta_2$ are fixed complex and $k \neq 1$.

Ans. Centre $\equiv \frac{k^2 \beta - \alpha}{k^2 - 1}$, Radius $= \frac{1}{(k^2 - 1)} \sqrt{|\alpha - k^2 \beta|^2 - (k^2 \cdot |\beta|^2 - |\alpha|^2) \cdot (k^2 - 1)}$

Sol. -

Q25. If one of the vertices of the square circumscribing the circle $|z - 1| = \sqrt{2}$ is $2 + \sqrt{3}i$. Find the other vertices of square.

Ans. $z_2 = -\sqrt{3}i$; $z_3 = (1 - \sqrt{3}) + i$; $z_4 = (1 + \sqrt{3}) - i$

Sol. -

Q26. Let $|z| = 2$ and $w = \frac{z + 1}{z - 1}$ where $z, w \in \mathbb{C}$ (where $\mathbb{C}$ is the set of complex numbers). If M and m respectively be the greatest and least modulus of w , then find the value of $(2010m + M)$.

Ans. -

Sol. Let $z = a + ib$ hence $a, b \in [-2, 2]$

$$w = \frac{(a + 1) + ib}{(a - 1) + ib}; \quad a^2 + b^2 = 4$$

$$|w| = \sqrt{\frac{(a + 1)^2 + b^2}{(a - 1)^2 + b^2}} = \sqrt{\frac{a^2 + b^2 + 2a + 1}{a^2 + b^2 - 2a + 1}} = \sqrt{\frac{5 + 2a}{5 - 2a}}$$

$$|w|_{\max} = \sqrt{\frac{5 + 4}{1}} = 3 = M, \quad \text{when } a = 2 \quad (z = 2)$$

$$|w|_{\min} = \sqrt{\frac{5 - 4}{9}} = \frac{1}{3} = m, \quad \text{when } a = -2 \quad (z = -2)$$

Hence $(2010m + M) = 673$ Ans.

Q27. If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the minimum value of $|2z - 6 + 5i|$ is _____.

Ans. 5

Sol. Given, $|z - 3 - 2i| \leq 2$ _____(1)

To find minimum of $|2z - 6 + 5i|$

$$\text{Or } 2 \left| z - 3 + \frac{5}{2}i \right|$$

using triangle inequality

$$\text{i.e., } ||z_1| - |z_2|| \leq |z_1 + \bar{z}_2|$$

$$\therefore \left| z - 3 + \frac{5}{2}i \right|$$

$$= \left| z - 3 - 2i + 2i + \frac{5}{2}i \right| = \left| (z - 3 - 2i) + \frac{9}{2}i \right| \geq \left| |z - 3 - 2i| - \frac{9}{2} \right| \geq \left| 2 - \frac{9}{2} \right|$$

$$\Rightarrow \left| z - 3 + \frac{5}{2}i \right| \geq \frac{5}{2} \quad \text{or } |2z - 6 + 5i| \geq 5$$

Q28. If Z is a non-real complex number, then find the minimum value of $\frac{imZ^5}{im^5Z}$.

Ans. -4

Sol. $-$

Q29. Find the sum of the series $1(2 - \omega)(2 - \omega^2) + 2(3 - \omega)(3 - \omega^2) \dots (n - 1)(n - \omega)(n - \omega^2)$ where ω is one of the imaginary cube root of unity.

Ans. $\left[\frac{n(n+1)}{2} \right]^2 - n$

Sol. $-$

Q30. Let ω be the complex number $\frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex

number z satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ is equal to

Ans. 1

Sol. $-$

Q31. If $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -3/2$ then prove that:

$$(a) \sum \cos 2\alpha = 0 = \sum \sin 2\alpha \quad (b) \sum \sin(\alpha + \beta) = 0 = \sum \cos(\alpha + \beta)$$

$$(c) \sum \sin^2 \alpha = \sum \cos^2 \alpha = 3/2 \quad (d) \sum \sin 3\alpha = 3 \sin(\alpha + \beta + \gamma)$$

$$(e) \sum \cos 3\alpha = 3 \cos(\alpha + \beta + \gamma)$$

$$(f) \cos^3(\theta + \alpha) + \cos^3(\theta + \beta) + \cos^3(\theta + \gamma) = 3 \cos(\theta + \alpha) \cdot \cos(\theta + \beta) \cdot \cos(\theta + \gamma)$$

where $\theta \in \mathbb{R}$.

Ans. $-$

Sol. $-$

Q32. Resolve $Z^5 + 1$ into linear & quadratic factors with real coefficients. Deduce that :

$$4 \cdot \sin \frac{\pi}{10} \cdot \cos \frac{\pi}{5} = 1.$$

Ans. $(Z + 1)(Z^2 - 2Z \cos 36^\circ + 1)(Z^2 - 2Z \cos 108^\circ + 1)$

Sol. $-$

Paragraph

For the complex number $w = \frac{4z - 5i}{2z + 1}$

\n

Q33. The locus of z , when w is a real number other than 2, is

(A) a point circle (B) a straight line with slope $-\frac{5}{2}$ and y-intercept $\frac{5}{4}$

(C) a straight line with slope $\frac{5}{2}$ and y-intercept $\frac{5}{4}$

(D) a straight line passing through the origin

Ans. C

Sol. -

Q34. The locus of z , when w is a purely imaginary number is

(A) a circle with centre $\left(\frac{1}{2}, -\frac{5}{4}\right)$ passing through origin.

(B) a circle with centre $\left(-\frac{1}{4}, \frac{5}{8}\right)$ passing through origin.

(C) a circle with centre $\left(\frac{1}{4}, -\frac{5}{8}\right)$ and $\frac{\sqrt{29}}{8}$ radius (D) any other circle

Ans. B

Sol. -

Q35. The locus of z , when $|w| = 1$ is

(A) a circle with centre $\left(-\frac{5}{8}, \frac{1}{4}\right)$ and radius $\frac{1}{2}$

(B) a circle with centre $\left(\frac{1}{4}, -\frac{5}{8}\right)$ and radius $\frac{1}{2}$

(C) a circle with centre $\left(\frac{5}{8}, -\frac{1}{4}\right)$ and radius $\frac{1}{2}$ (D) any other circle

Ans. D

Sol. -

Paragraph

Let A, B, C be three sets of complex numbers as defined below.

$$A = \{z : |z + 1| \leq 2 + \operatorname{Re}(z)\}, \quad B = \{z : |z - 1| \geq 1\} \quad \text{and} \quad C = \left\{z : \left|\frac{z-1}{z+1}\right| \geq 1\right\}$$

\n

Q36. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is

(A) 4 (B) 5 (C) 6 (D) 10

Ans. B

Sol. -

Q37. The area of region bounded by $A \cap B \cap C$ is

(A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $4\sqrt{3}$ (D) 2

Ans. A

Sol. -

Q38. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude is

- (A) -1 (B) $\frac{-3}{2}$ (C) $\frac{1}{2}$ (D) -2

Ans. B

Sol. -

Q39. If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$, then the value of 'r' can be :

- (A) $\cot \frac{\pi}{8}$ (B) $\sec \pi$ (C) $\tan \frac{\pi}{12}$ (D) $\tan \frac{5\pi}{12}$

Ans. B, C, D

Sol. $(1 + ir)^3 = 1^3 + 3ir + 3(ir)^2 + (ir)^3$
 $= 1 + i(3r) + 3(-1)r^2 - r^3i = s + si$
 $= (1 - 3r^2) + i(3r - r^3) = s + si$
 $= 1 - 3r^2 = s$ and $3r - r^3 = s$
 $1 - 3r^2 = 3r - r^3$
 $1 = \frac{3r - r^3}{1 - 3r^2}$
 $1 = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = \tan 3\theta$
 $3\theta = \frac{\pi}{4}$ $r = \tan \theta$
 $\theta = \frac{\pi}{12}$, $r = \tan \frac{\pi}{12}$

Q40. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t with

$0 < t < 1$. If $\text{Arg}(w)$ denotes the principal argument of a nonzero complex number w , then

- (A) $|z - z_1| + |z - z_2| = |z_1 - z_2|$ (B) $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$
 (C) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z} \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$ (D) $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$

Ans. A, C, D

Sol. -

- Q41. POQ is a straight line through the origin O . P and Q represent the complex number $a + ib$ and $c + id$ respectively and $OP = OQ$. Then
- (A) $|a + ib| = |c + id|$ (B) $a + c = b + d$ (C) $\arg(a + ib) = \arg(c + id)$
- (D) none of these

Ans. A, B

Sol. -

- Q42. If z satisfies the inequality $|z - 1 - 2i| \leq 1$, then
- (A) $\min(\arg(z)) = \tan^{-1}\left(\frac{3}{4}\right)$ (B) $\max(\arg(z)) = \frac{\pi}{2}$ (C) $\min(|z|) = \sqrt{5} - 1$
- (D) $\max(|z|) = \sqrt{5} + 1$

Ans. A, B, C, D

Sol. -

- Q43. If $|z_1| = |z_2| = |z_3| = 1$ and z_1, z_2, z_3 are represented by the vertices of an equilateral triangle then
- (A) $z_1 + z_2 + z_3 = 0$ (B) $z_1 z_2 z_3 = 1$ (C) $z_1 z_2 + z_2 z_3 + z_3 z_1 = 0$ (D) none of these

Ans. A, C

Sol. -

- Q44. Let A and B be two distinct points denoting the complex numbers α and β respectively. A complex number z lies between A and B where $z \neq \alpha, z \neq \beta$. Which of the following relation(s) hold good?

- (A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$
- (B) $\exists$ a positive real number 't' such that $z = (1 - t)\alpha + t\beta$ (C) $\begin{vmatrix} z - \alpha & \bar{z} - \bar{\alpha} \\ \beta - \alpha & \bar{\beta} - \bar{\alpha} \end{vmatrix} = 0$
- (D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

Ans. A, B, C, D

Sol. -

- Q45. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$. Which of the following statement(s) is/are correct?

- (A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

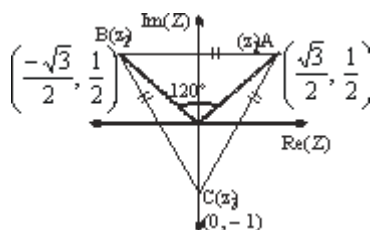
$$(B) \sum_{k=1}^3 \text{Amp}(z_k) = \frac{\pi}{2}$$

$$(C) \text{ Centroid of the triangle formed by } z_1, z_2 \text{ and } z_3 \text{ is } \left(\frac{1}{\sqrt{3}}, \frac{-1}{3} \right)$$

$$(D) \text{ Area of triangle formed by } z_1, z_2 \text{ and } z_3 \text{ is } \frac{3\sqrt{3}}{2}$$

Ans. A, B

Sol.



$$\text{We have } z^4 = iz \Rightarrow z^3 = i$$

$$\Rightarrow z = e^{i(4k+1)\frac{\pi}{6}} \text{ (Using D.M.T.)}$$

Put $k = 0, 1, 2$, we get

$$z_1 = e^{i\frac{\pi}{6}}, z_2 = e^{i\frac{5\pi}{6}} \text{ and } z_3 = e^{i\frac{3\pi}{2}}$$

Clearly triangle formed by z_1, z_2 and z_3 is equilateral.

$$\therefore \text{ centroid of } \triangle ABC \text{ is } (0, 0) \text{ and Area } (\triangle ABC) = \frac{3\sqrt{3}}{4}$$

Q46. If $z \in \mathbb{C}$, which of the following relation(s) represents a circle on an Argand diagram?

$$(A) |z - 1| + |z + 1| = 3 \quad (B) (z - 3 + i)(\bar{z} - 3 - i) = 5 \quad (C) 3|z - 2 + i| = 7$$

$$(D) |z - 3| = 2$$

Ans. B, C, D

Sol. (A) is obviously ellipse

$$(B) (z - \alpha) = 5 \text{ where } \alpha = 3 - i; \bar{\alpha} = 3 + i$$

$$|z - \alpha|^2 = 5 \Rightarrow |z - \alpha| = \sqrt{5} \text{ circle with centre } (3, -1) \text{ and radius } = \sqrt{5} \Rightarrow (B) \text{ is correct}$$

$$(C) |z - (2 - i)| = \frac{7}{3} \Rightarrow \text{circle with centre } (2, -1) \text{ and radius } = \frac{7}{3} \Rightarrow (C) \text{ is correct}$$

$$(D) |z - 3| = 2 \Rightarrow \text{circle with centre } (3, 0) \text{ and radius } = 2 \Rightarrow (D) \text{ is correct}$$

Q47. Equation of a straight line on the complex plane passing through a point P denoting the complex

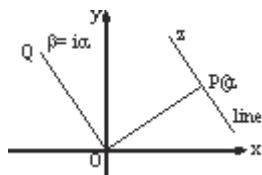
number a and perpendicular to the vector $\overrightarrow{OP}$ where 'O' is the origin can be written as :

$$(A) \operatorname{Im} \left(\frac{z - \alpha}{\alpha} \right) = 0 \quad (B) \operatorname{Re} \left(\frac{z - \alpha}{\alpha} \right) = 0 \quad (C) \operatorname{Re}(\bar{\alpha} z) = 0$$

$$(D) \bar{\alpha} z + \alpha \bar{z} - 2|\alpha|^2 = 0$$

Ans. B, D

Sol.



Required line is passing through $P(\alpha)$ and parallel to the vector $\overrightarrow{OQ}$

hence $z = \alpha + i\lambda\alpha, \lambda \in \mathbb{R}$

$$\frac{z - \alpha}{\alpha} = \text{purely imaginary}$$

$$\operatorname{Re} \left(\frac{z - \alpha}{\alpha} \right) = 0 \Rightarrow (B) \quad (\text{multiply } N^r \text{ and } D^r \text{ by } \bar{\alpha})$$

$$\operatorname{Re}((z - \alpha)\bar{\alpha}) = 0 \Rightarrow \operatorname{Re}(z\bar{\alpha} - |\alpha|^2) = 0$$

$$\text{also } \frac{z - \alpha}{\alpha} + \frac{\bar{z} - \bar{\alpha}}{\bar{\alpha}} = 0$$

$$\bar{\alpha}(z - \alpha) + \alpha(\bar{z} - \bar{\alpha}) = 0$$

$$\bar{\alpha}z + \alpha\bar{z} - 2|\alpha|^2 = 0 \Rightarrow (D)$$

Q48. Match the statements in Column I with those in Column-II

[Note : Here z takes values in the complex plane and $\operatorname{Im} z$ and $\operatorname{Re} z$ denote, respectively, the imaginary part and the real part of z .]

Column I

Column II

(A) The set of points z satisfying

$$|z - i|z| = |z + i|z|$$

is contained in or equal to

(B) The set of points z satisfying

$$|z + 4| + |z - 4| = 10$$

is contained in or equal to

(C) If $|w| = 2$, then the set of points

$z = w - \frac{1}{w}$ is contained in or equal to

(p) an ellipse with eccentricity $\frac{4}{5}$

(q) the set of points z satisfying $\operatorname{Im} z = 0$

(r) the set of points z satisfying $|\operatorname{Im} z| \leq 1$

- (D) If $|w| = 1$, then the set of points $z =$ is contained in or equal to (s) the set of points z satisfying $|\operatorname{Re} z| \leq 2$
 (t) the set of points z satisfying $|z| \leq 3$
- (A) (A) q, r (B) p (C) p,s,t (D) q,r,s,t (B) (A) q, r (B) r (C) p,r,t (D) q,r,s,t
- (C) (A) q, (B) p (C) p,s,t (D) q,r (D) (A) p, s (B) r (C) p,r,t (D) q,r,s,t

Ans. 5ef09cde4c298c76832c044b

Sol. -

Q49. Match the column

Column-I

Column-II

- (A) Let w be a non real cube root of unity then the number of distinct elements (P) 4
 in the set $\{1 + \omega + \omega^2 + \dots + \omega^n\}^m | m, n \in N$ is
- (B) Let $1, \omega, \omega^2$ be the cube root of unity. The least possible (Q) 5
 degree of a polynomial with real coefficients having roots
 $2\omega, (2 + 3\omega), (2 + 3\omega^2), (2 - \omega - \omega^2)$, is
- (C) $\alpha = 6 + 4i$ and $\beta = (2 + 4i)$ are two complex numbers on the complex plane. (R) 6
 A complex number z satisfying $\operatorname{amp} \left(\frac{z - \alpha}{z - \beta} \right) = \frac{\pi}{6}$ moves on the major
- (S) 7
 segment of a circle whose radius is
- (A) (A) – S (B) – R (C) – Q (B) (A) – S (B) – Q (C) – P
- (C) (A) – P (B) – S (C) – Q (D) (A) – R (B) – Q (C) – P

Ans. 5ef074df4c298c76832b7f98

Sol. -

Q50. Match the column

Let z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$.

Column – I

Column – II

- (A) Maximum value of $|z_1 + z_2|$ (p) 3
 (B) Minimum value of $|z_1 - z_2|$ (q) 1
 (C) Minimum value of $|2z_1 + 3z_2|$ (r) 4
 (D) Maximum value of $|z_1 - 2z_2|$ (s) 5

- (A) (A) – p (B) – q (C) – r (D) – s (B) (A) – s (B) – q (C) – r (D) – p

$$^{(C)} (A) - p (B) - r (C) - q (D) - s \quad ^{(D)} (A) - q (B) - p (C) - s (D) - r$$

Ans. 5ef07663239ecc768b155f4d

Sol. -